

(2½ Hours)

[Total Marks: 75]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculators is **allowed**.

1. Attempt any three of the following:**15**

- What is Java Enterprise Edition (Java EE)? Explain.
- Write a note on Multi Tier EE application architecture.
- List and explain the tasks that Servlet can do.
- Explain the life cycle of servlet.
- Write a servlet code to display Square and Square root of numbers between 25 and 365 in tabular form.
- List and explain four types of JDBC drivers.

2. Attempt any three of the following:**15**

- Explain Cookie class with its constructor and any five methods.
- Write a servlet program to create a session and display the following:
 i) Session ID ii) New or Old iii) Creation Time
- Write a servlet program **GradeServlet.java** that accepts Grade through radio buttons from **index.html** page, if the string is “A”, “B”, “C” OR “D”, the program should dispatch the direct to the **Success.html** page containing message “**Congratulations !!!, You Passed SEM V exam**”, else display “**Try Again**” and load index.html page in current servlet. **(Make suitable assumptions, Only Write the Servlet Code)**
- Explain the following w. r. t. working with files in Servlet.
 i) @MultipartConfigure
 ii) fileSizeThreshold
 iii) location
 iv) maxFileSize
 v) maxRequestSize
- Explain using a code snippet the onDataAvailable() and onAllDataRead() methods of ReadListener interface.
- What is RequestDispatcher? Explain the two methods of RequestDispatcher.

3. Attempt any three of the following:**15**

- Explain the reasons to use JSP.
- What are directives? Explain page directive with any of its four attributes.
- Develop a simple JSP application to accept values from html page and display on next page. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).
- Explain the <jsp:useBean > tag with its attribute. Support your answer with suitable code snippet.
- List the name of JSP implicit objects. Explain any Two in details.
- What is wrong in using JSP scriptlet tag? How JSTL fixes JSP scriptlet shortcomings?

4. Attempt any three of the following:**15**

- Explain benefits of EJB.
- Write a note on different types of session beans.
- Explain life cycle of a message driven bean using suitable diagram.
- Write a stateless session bean code to represent BookInformation. (BookId integer, BookName String, Pages integer, Price double)
- What is an interceptor? How an interceptor is defined and how aroundInvoke () is added to it?
- What is Java Naming and Directory Interface? Explain.

5. Attempt any three of the following:**15**

- What is Persistence? Explain.
- Explain with suitable diagram architecture of Java Persistence API.
- Write a JSP code to add guest feedback using JPA in GuestBook table in database. (Make suitable assumptions)
- What is Hibernate? Explain Object Relational Mapping.
- Explain the architecture of Hibernate with suitable diagram.
- Write a JSP code to add visitor's feedback using Hibernate in FeedBack table in database. (Make suitable assumptions)